MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
MG5DSCCHE301 - PHYSICAL CHEMISTRY – 2
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions, Answer all Questions.
Each question carries 1 mark.

1. Crystalline solids generally exhibit [U, CO= 1]
   A. Isotropy
   B. Anisotropy
   C. No definite geometry
   D. Irregular cleavage
2. A plane cuts the crystallographic axis at 2a,3b and 6c. The Miller indices of the plane are [U, CO= 1]

   A.(236)
   B.(632)
   C.(321)
   D. (123)
3. In the CsCl structure, the coordination number of Cs^+ and Cl^- ions is [U, CO= 1]
   A. 4:4
   B. 6:6
   C. 8:8
   D. 12:12
4. According to the Stark–Einstein law, absorption of one photon by a molecule generally results in: [K, CO= 2]
   A. Excitation of one molecule
   B. Excitation of two molecules
   C. Ionization of all molecules
   D. Emission of two photons
5. Fluorescence generally involves emission from: [U, CO= 2]
   A. $S_0 \rightarrow S_1$
   B. $S_1 \rightarrow S_0$
   C. $T_1 \rightarrow S_0$
   D. $T_1 \rightarrow T_2$

6. Which of the following is a typical example of a photochemical reaction having a high quantum yield?

[U, CO= 2]

   A. Photochemical decomposition of HBr
   B. Fluorescence of an aromatic molecule
   C. Internal conversion
   D. Phosphorescence

7. The relation between enthalpy and internal energy is [U, CO= 3]
   A. H=U-PV
   B. H=U+PV
   C. H=PV-U
   D. H=U/PV
8. A spontaneous process at constant temperature and pressure is characterized by [U, CO= 3]
   A. $\Delta G>0$
   B. $\Delta G<0$
   C. $\Delta G=0$
   D. $\Delta H=0$
9. Chemical potential is defined as the change in Gibbs free energy with respect to [U, CO= 3]
   A. Temperature
   B. Pressure
   C. Volume
   D. Number of moles
10. The efficiency of a Carnot engine depends on [U, CO= 3]
   A. Working substance
   B. Pressure only
   C. Temperature of source and sink
   D. volume only

[1 × 10 = 10]

Part B

Short Answer Type Questions, Answer any four Questions
Each question carries 5 marks

11. Derive the Bragg's equation. [U, CO= 1]
12. Discuss the phenomenon of superconductivity. Discuss its characteristic properties and applications. [U, CO= 1]
13. Give examples of photochemical reactions with high quantum yield and low quantum yield. [U, CO= 2]
14. Illustrate the statement, "Chemiluminescence is regarded as the reverse of a photochemical reaction." [U, CO= 2]
15. State and explain the different statements of the second law of thermodynamics. [U, CO= 3]
16. Derive the expression for entropy as a function of volume and temperature. [A, CO= 3]

[4 × 5 = 20]

Part C

Long Essay Type Questions, Answer any Two Questions
Each question carries 10 marks

17. Explain the different types of stoichiometric defects found in crystals. [U, CO= 1]
18. Define entropy and explain its physical significance. Derive the entropy change of an ideal gas as a function of volume and temperature, and as a function of pressure and temperature. [U, CO= 3]
19. Derive expressions for work done during reversible isothermal and adiabatic expansion of an ideal gas. [U, CO= 3]
20. Define heat capacities at constant volume and constant pressure and derive their relationship for an ideal gas. [U, CO= 3]

[2 × 10 = 20]
